

BOGUE USMC, NC, USA

WMO: 723097

Lat: 34.683N

Lon: 77.033W

Elev: 6

StdP: 101.25

Time Zone: -5.00 (NAE)

Period: 07-19

WBAN: 93743

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -4.8	(c) -2.7	(d) -15.0	(e) 1.0	(f) -1.7	(g) -12.2	(h) 1.3	(i) 0.4	(j) 9.7	(k) 17.3	(l) 8.5	(m) 12.6	(n) 2.7	(o) 340	(p) 0.432

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 6.4	(c) 32.0	(d) 25.4	(e) 31.0	(f) 25.1	(g) 30.2	(h) 24.8	(i) 26.9	(j) 30.0	(k) 26.5	(l) 29.7	(m) 26.0	(n) 29.0	(o) 4.8	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 26.1	(b) 21.4	(c) 28.9	(d) 25.6	(e) 20.8	(f) 28.5	(g) 25.1	(h) 20.2	(i) 28.2	(j) 84.7	(k) 30.2	(l) 82.4	(m) 29.8	(n) 80.5	(o) 29.1	(p) 28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 8.8	(b) 7.9	(c) 7.2	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.6	7.5	9.7	12.4	16.8	21.6	25.1	26.4	25.1	24.0	19.1	12.5	10.2
	DBStd	7.81	6.17	5.95	4.72	3.71	3.16	2.39	2.60	3.46	2.83	4.29	4.64	4.58
	HDD10.0	309	124	71	29	2	0	0	0	0	0	1	26	57
	HDD18.3	1342	340	248	189	69	10	0	0	4	2	44	181	254
	CDD10.0	3070	45	63	104	207	359	453	507	467	419	282	102	62
	CDD18.3	1063	3	7	6	24	111	204	249	213	171	67	7	1
	CDH23.3	8883	38	76	37	45	551	1777	2599	2038	1314	390	14	5
	CDH26.7	2364	11	18	7	3	85	429	852	613	298	46	0	1
Wind	WSAvg	3.4	3.0	3.3	3.6	3.9	3.9	3.8	3.7	3.1	3.3	3.1	3.0	3.0
Precipitation	PrecAvg	1410	103	94	103	81	104	132	163	181	166	100	87	97
	PrecMax	2005	199	259	252	173	247	285	299	370	495	268	233	184
	PrecMin	940	45	42	40	14	44	54	84	67	27	2	18	28
	PrecStd	243	42	49	47	39	48	58	60	70	111	67	57	39
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	28.2	29.5	27.5	26.0	31.7	33.1	33.9	33.0	31.4	29.6	25.0	22.5
		MCWB	18.8	18.0	17.8	21.3	23.0	24.2	25.6	25.2	23.9	23.8	21.1	17.7
	2%	DB	22.7	25.6	23.2	24.6	28.5	31.2	31.9	31.3	30.2	27.9	23.2	20.3
		MCWB	15.8	16.3	16.4	20.0	22.7	24.3	26.1	25.5	24.4	23.6	20.0	17.8
	5%	DB	19.1	23.0	21.3	23.5	27.2	30.0	31.0	30.4	29.4	26.8	21.8	19.2
		MCWB	15.1	16.7	16.5	19.2	22.4	24.2	25.8	25.1	24.3	22.7	19.0	17.2
	10%	DB	16.9	19.0	19.5	22.5	26.1	29.0	30.2	29.5	28.5	25.6	20.3	17.9
		MCWB	14.4	14.8	15.8	18.6	21.7	23.8	25.4	24.8	23.8	22.0	17.5	16.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.4	20.6	19.7	23.0	25.1	26.8	27.9	27.3	26.5	25.6	22.3	19.6
		MCDB	26.3	26.3	23.0	24.6	28.3	29.3	31.4	29.9	29.3	27.4	24.0	21.3
	2%	WB	17.7	18.7	18.6	21.4	24.0	26.1	27.2	26.7	25.7	24.7	21.1	18.4
		MCDB	20.6	22.2	21.2	23.4	27.3	29.2	30.5	29.7	28.7	26.7	22.7	19.7
	5%	WB	16.4	17.1	17.7	20.3	23.0	25.6	26.8	26.1	25.2	24.0	19.8	17.6
		MCDB	18.7	21.3	20.3	22.6	25.8	28.4	29.9	29.3	28.3	26.2	21.3	18.8
	10%	WB	14.7	15.5	16.6	19.4	22.5	25.1	26.4	25.7	24.6	22.9	18.1	16.4
		MCDB	16.6	18.8	19.1	21.6	25.1	27.8	29.4	28.7	27.6	25.2	19.8	17.8
Mean Daily Temperature Range	5% DB	MDBR	10.6	11.0	9.9	8.7	7.7	7.1	6.4	6.5	7.5	9.0	10.3	9.6
		MCDBR	13.2	15.8	10.6	8.3	7.5	7.8	7.0	6.8	7.4	8.1	10.4	9.4
		MCWBR	9.1	8.9	6.6	4.9	3.7	3.9	3.6	3.5	3.5	4.6	7.3	7.7
	5% WB	MCDBR	11.4	12.1	8.8	6.5	6.5	6.8	6.3	5.8	6.2	6.1	8.7	8.9
		MCWBR	9.2	8.4	6.6	4.6	3.6	4.0	3.7	3.5	3.6	4.2	7.6	8.2
Clear-Sky Solar Irradiance	taub	0.325	0.334	0.362	0.401	0.447	0.491	0.520	0.501	0.423	0.377	0.346	0.333	
	taud	2.534	2.504	2.444	2.331	2.247	2.169	2.102	2.149	2.371	2.476	2.511	2.538	
	Ebn at Noon	884	913	911	886	845	804	778	786	837	854	849	851	
	Edh at Noon	81	93	107	126	139	150	159	149	114	95	83	77	
All-Sky Solar Radiation	RadAvg	2.62	3.35	4.43	5.59	6.16	6.38	6.02	5.48	4.54	3.86	2.90	2.32	
	RadStd	0.16	0.32	0.34	0.37	0.42	0.35	0.36	0.32	0.30	0.40	0.21	0.17	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (35 neighbors)	+0.47	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+167	+118

Nomenclature: See separate page